



US012388050B2

(12) **United States Patent**
Leslie et al.

(10) **Patent No.:** **US 12,388,050 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **APPARATUSES AND METHODS FOR
COUPLING A PLURALITY OF
SEMICONDUCTOR DEVICES**

(58) **Field of Classification Search**
CPC H01L 2225/06562
See application file for complete search history.

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 31 days.

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(21) Appl. No.: **18/499,087**

(22) Filed: **Oct. 31, 2023**

(65) **Prior Publication Data**
US 2024/0063188 A1 Feb. 22, 2024

Related U.S. Application Data

(62) Division of application No. 17/349,657, filed on Jun.
16, 2021, now Pat. No. 11,837,580.

(60) Provisional application No. 63/039,880, filed on Jun.
16, 2020.

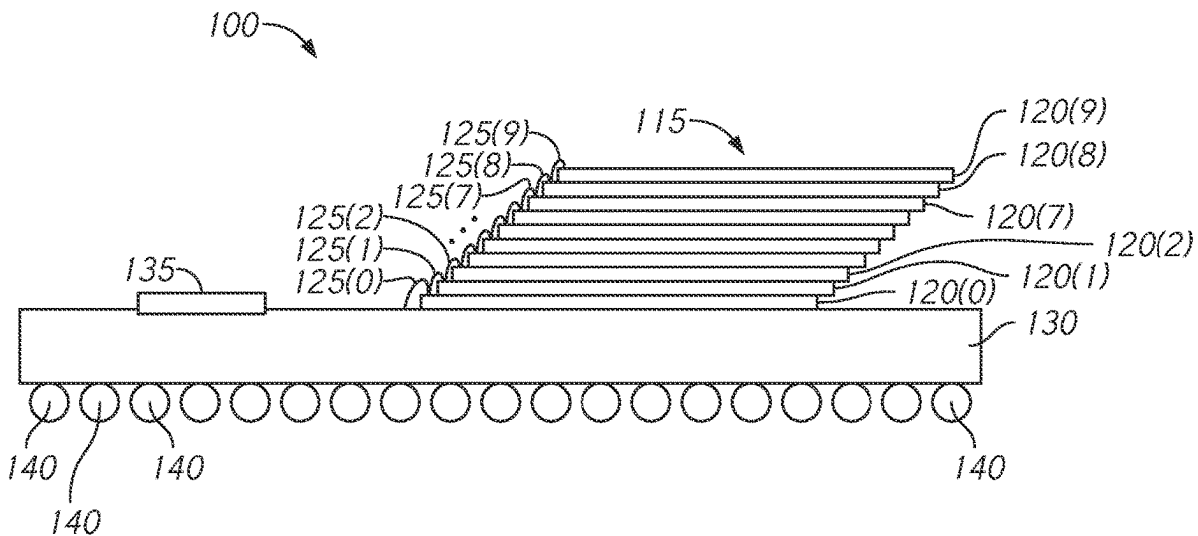
(51) **Int. Cl.**
H01L 25/065 (2023.01)
H01L 25/18 (2023.01)

(52) **U.S. Cl.**
CPC **H01L 25/0657** (2013.01); **H01L 25/18**
(2013.01); **H01L 2225/06506** (2013.01); **H01L**
2225/0651 (2013.01); **H01L 2225/06562**
(2013.01)

(57) **ABSTRACT**

Apparatuses and methods for coupling semiconductor devices are disclosed. In a group of semiconductor devices (e.g., a stack of semiconductor devices), a signal is provided to a point of coupling at an intermediate semiconductor device of the group, and the signal is propagated away from the point of coupling over different (e.g., opposite) signal paths to other semiconductor devices of the group. Loading from the point of coupling at the intermediate semiconductor device to other semiconductor devices of a group may be more balanced than, for example, having a point of coupling at semiconductor device at an end of the group (e.g., a lowest semiconductor device of a stack, a highest semiconductor device of the stack, etc.) and providing a signal therefrom. The more balanced topology may reduce a timing difference between when signals arrive at each of the semiconductor devices.

18 Claims, 8 Drawing Sheets



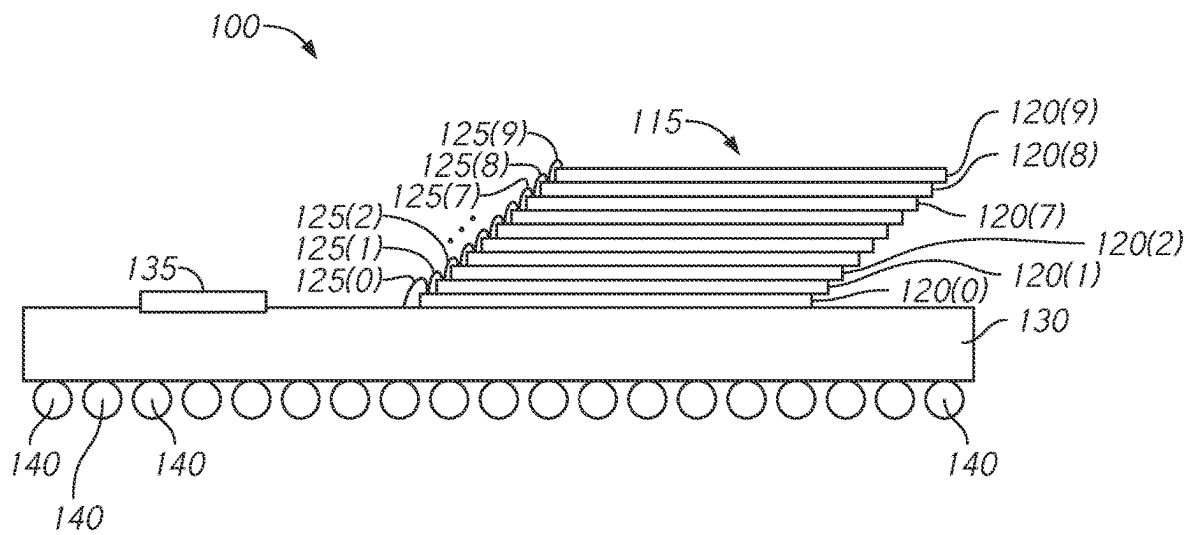


FIG. 1

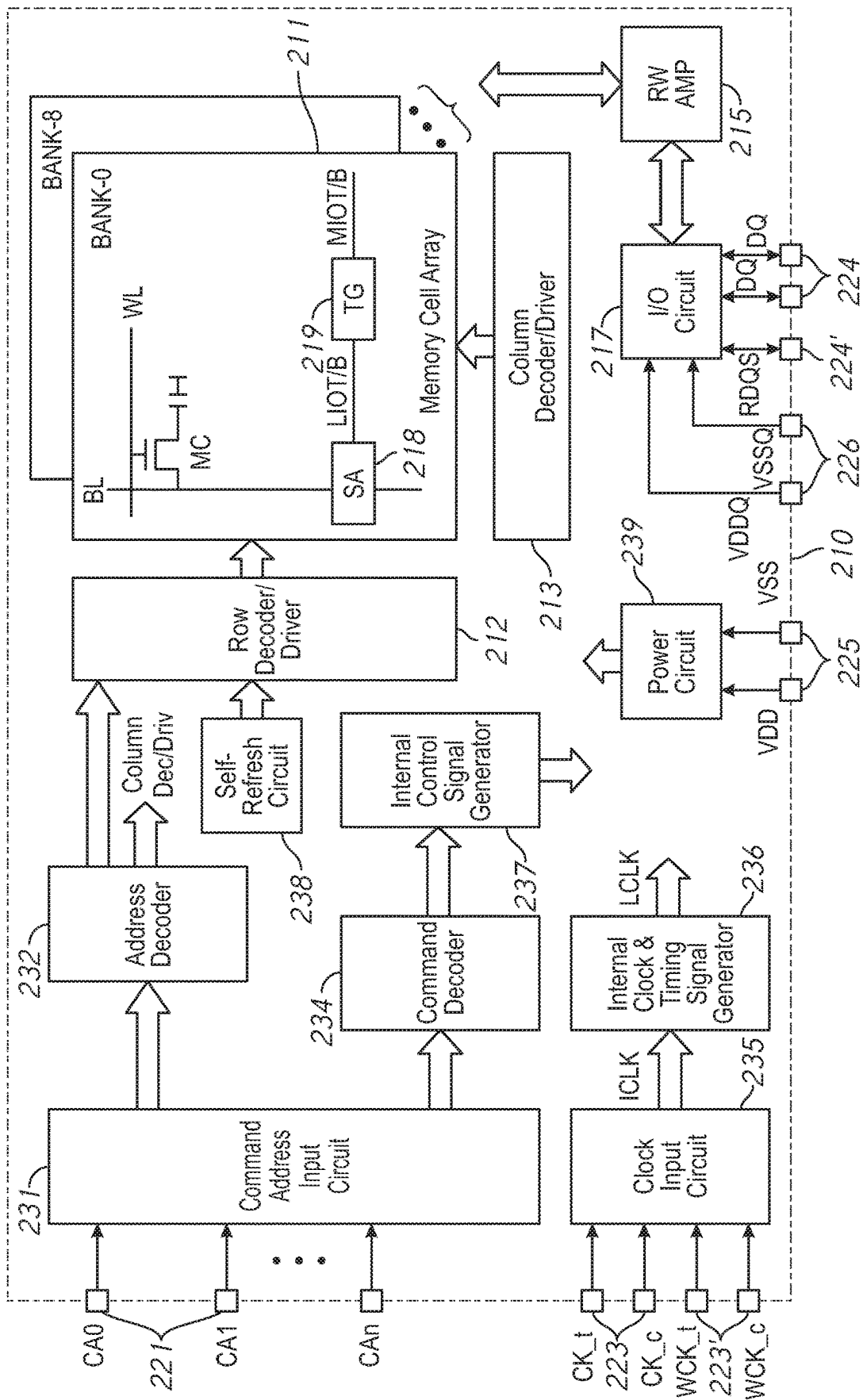
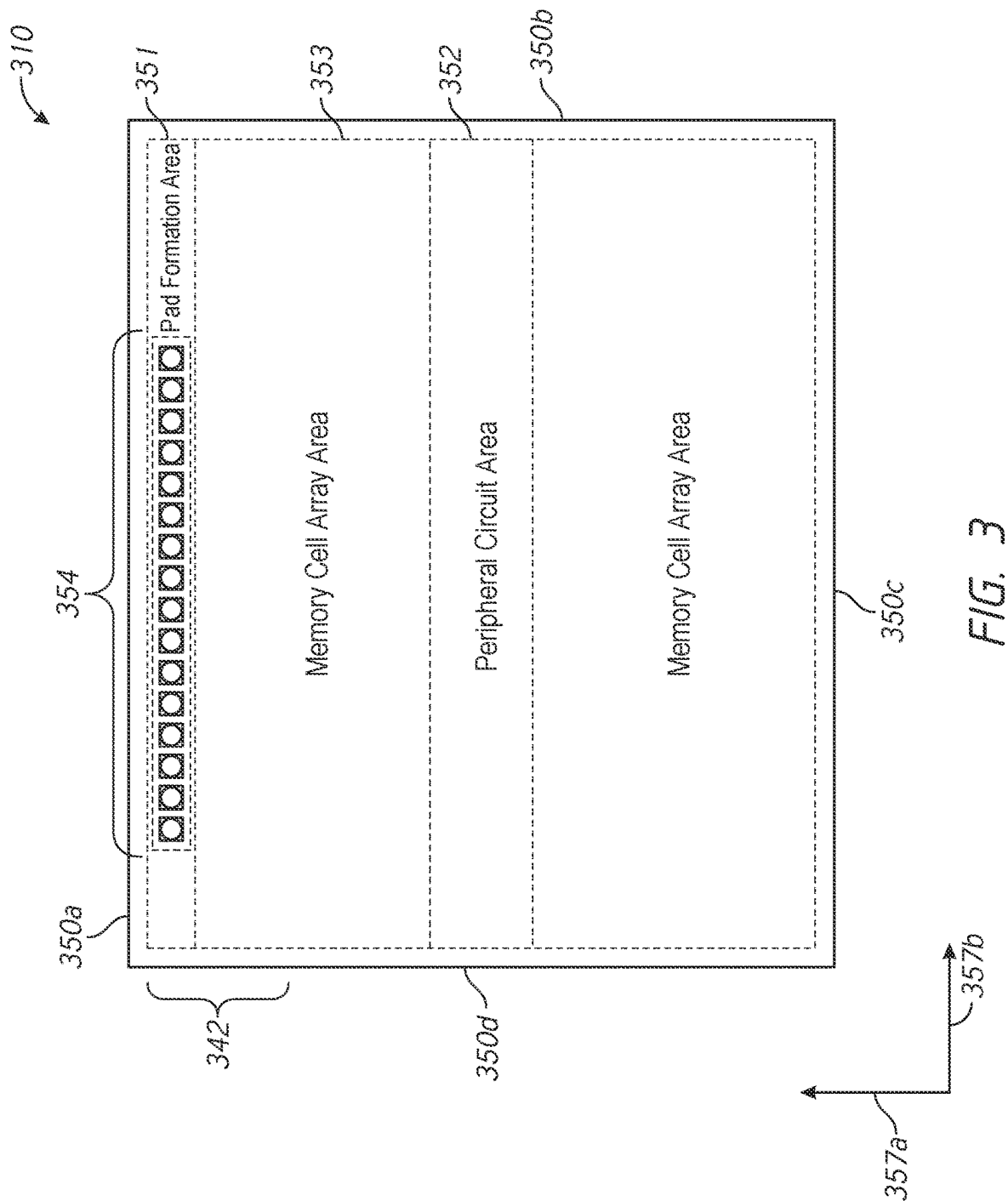


FIG. 2



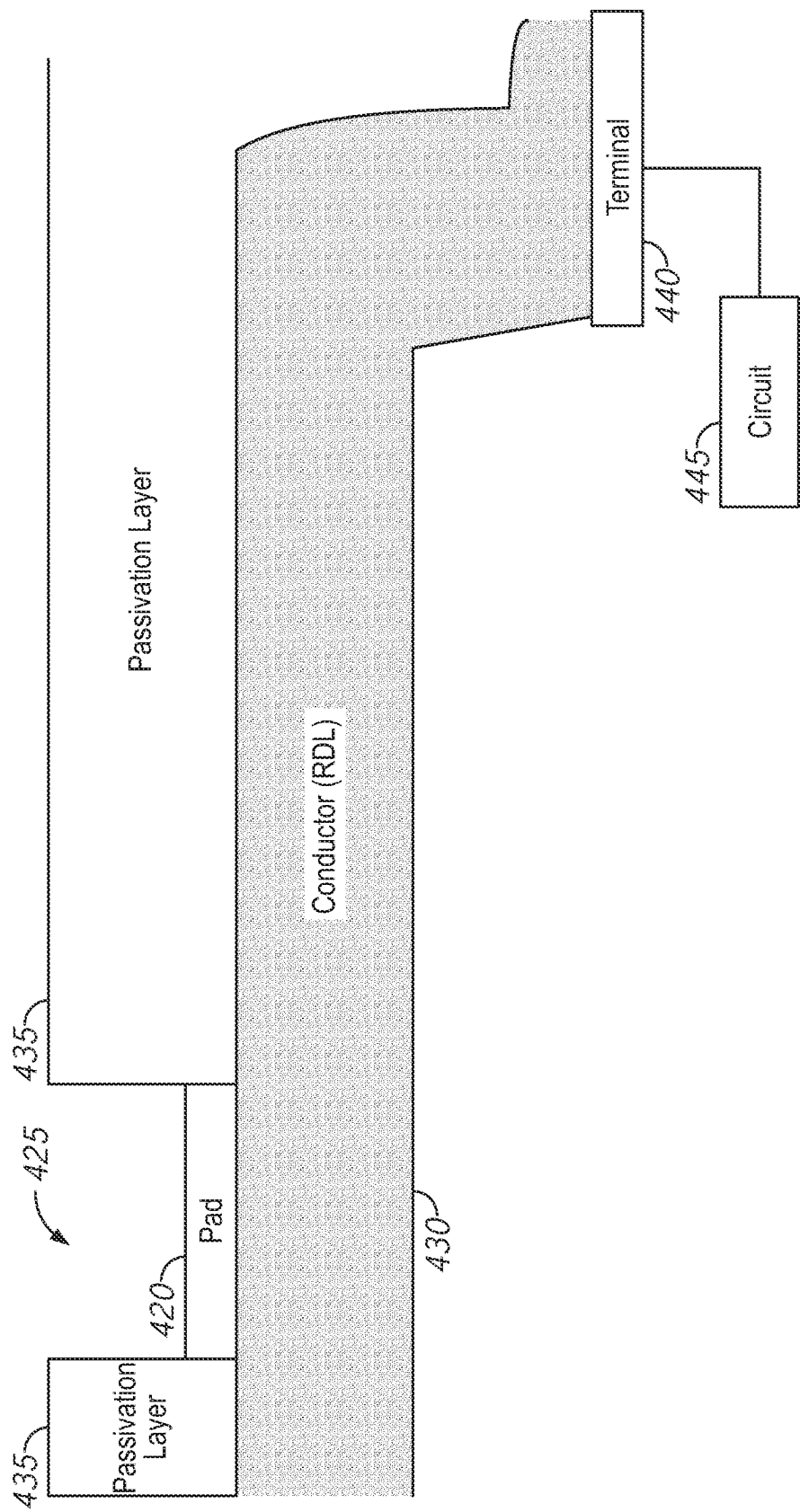


FIG. 4

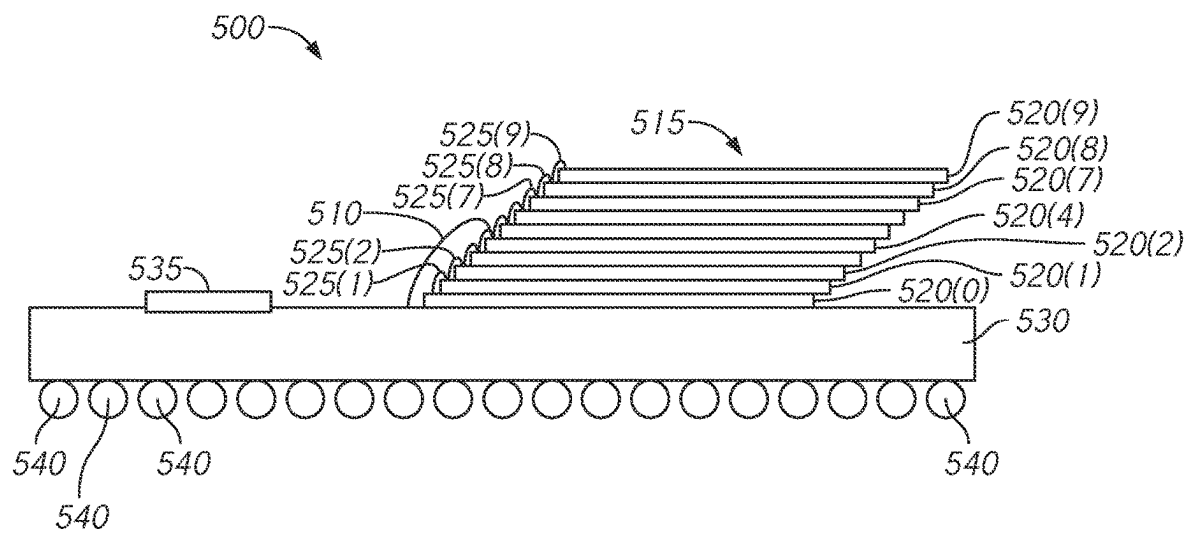


FIG. 5

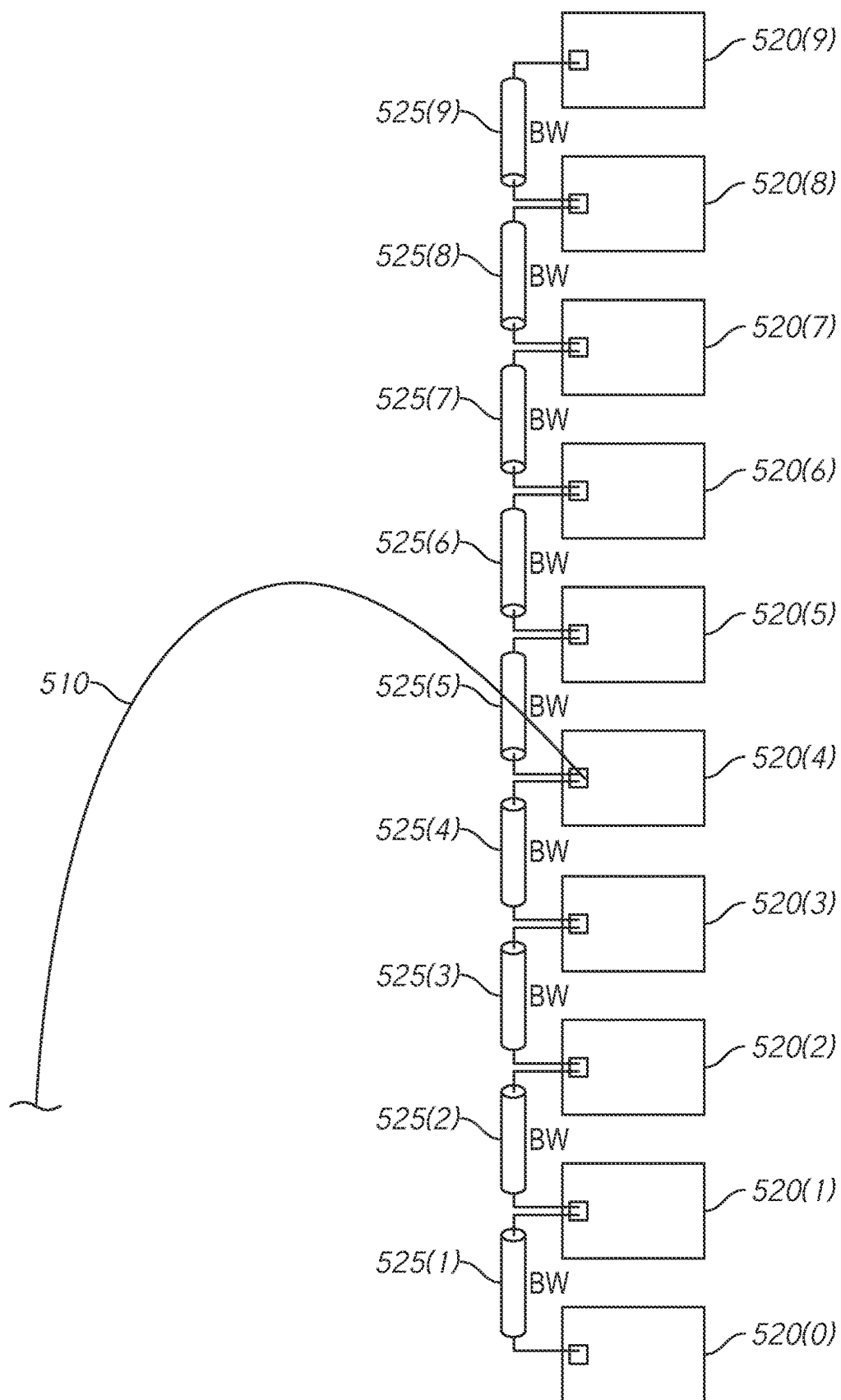


FIG. 6

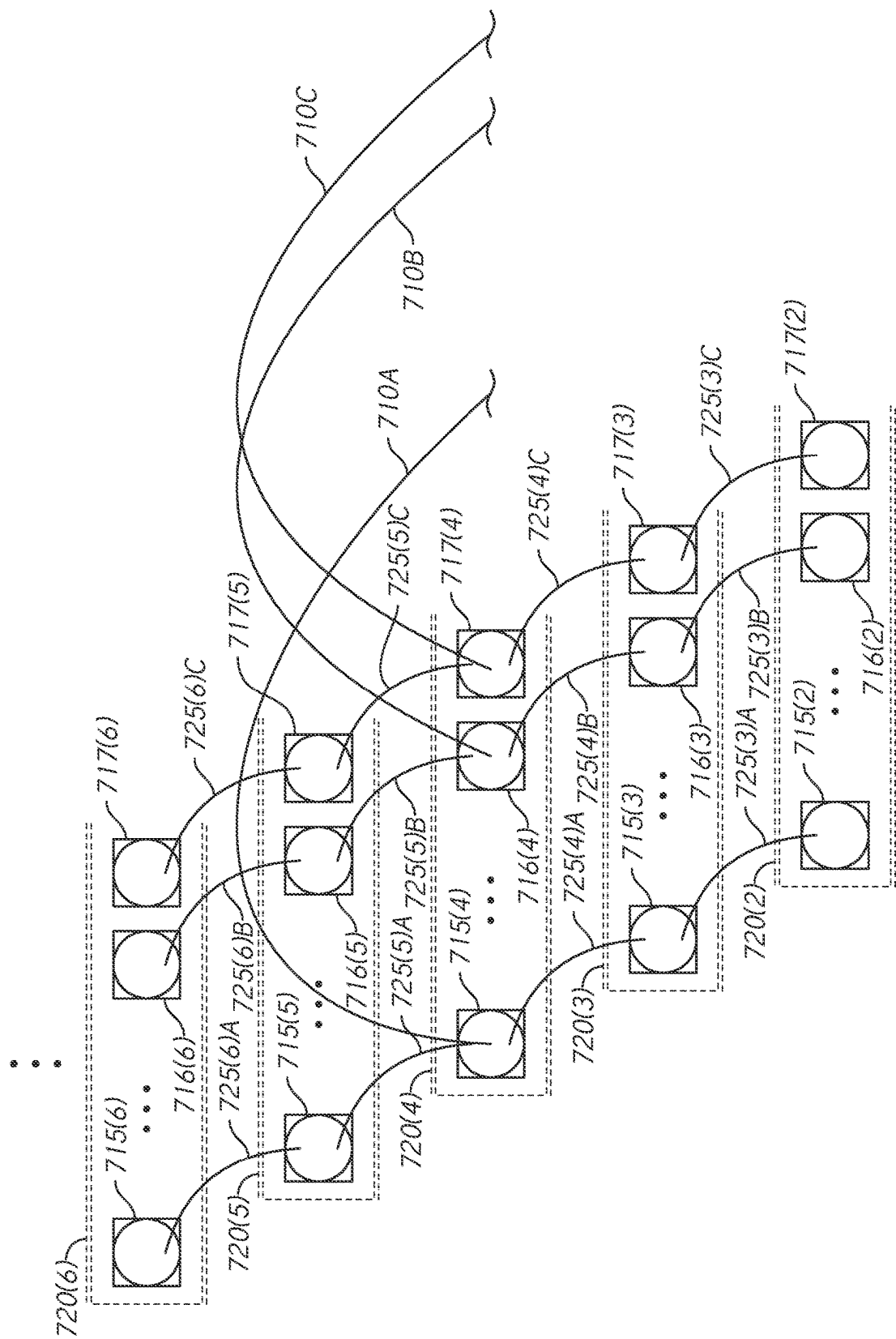
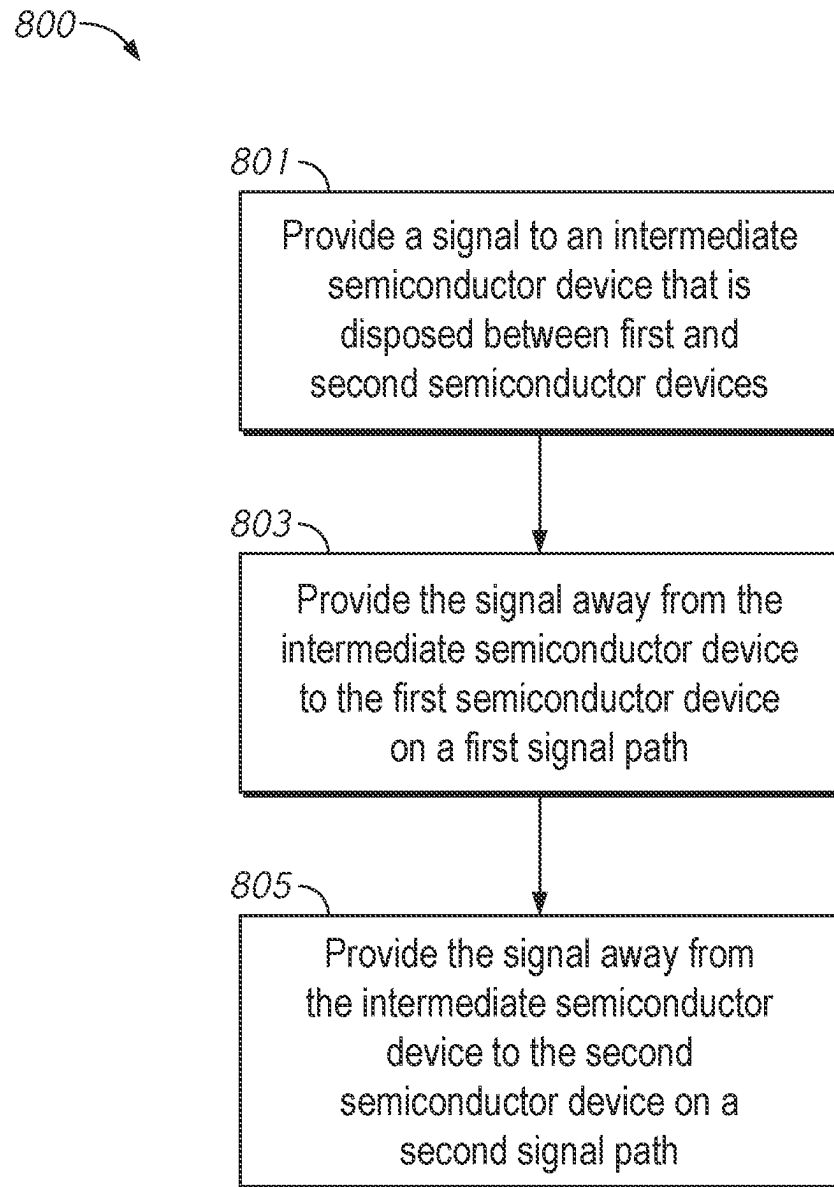


FIG. 7

*FIG. 8*

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APPARATUSES AND METHODS FOR COUPLING A PLURALITY OF SEMICONDUCTOR DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a divisional of U.S. patent application Ser. No. 17/349,657, filed Jun. 16, 2021, which claims the filing benefit of U.S. Provisional Application No. 63/039,880, filed Jun. 16, 2020. Both of these applications are incorporated by reference herein in their entireties and for all purposes.

BACKGROUND

In recent years, three-dimensional (3D) memory devices have been introduced. Some 3D memory devices are formed by stacking die vertically and coupling the die using through-silicon (or through-substrate) vias (TSVs) and/or wire bonds. Thus, 3D memory may also be referred to as “stacked memory.” 3D memory may provide greater memory capacity and/or higher bandwidth with less increase in area than non-3D memory. Example 3D memory devices include Hybrid Memory Cube (HMC), High Bandwidth Memory (HBM), and Master-Slave Memory (MSM), each of which may include multiple dynamic random access memory (DRAM) die coupled to one another in a stack.

The memory device may be coupled to an external circuit that provides command, address, and data signals to the stacked die to access the memory. The stacked dies are typically coupled the external circuit at the lowest elevation die, with cascade bonds connected upwards to the rest of the die in the stack.

As memory speeds increase, however, the delay between signals arriving at the lowest elevation die and the highest elevation die in the stack becomes non-negligible. The difference in timing of the arriving signals is typically accommodated by setting operational timing of the memory device based on the worst case, that is, the signal arrival time at the highest elevation die in the stack. This approach may result in reduced performance for the memory device. Additionally, as memory speeds continue to increase, the timing difference in signal arrival time from a lowest die to a highest die in the stack may no longer be tolerable for desired system performance.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram showing a multi-die device.

FIG. 2 is a block diagram of a semiconductor device in accordance with an embodiment of the present disclosure.

FIG. 3 is a layout diagram of a semiconductor device in accordance with one embodiment of the present disclosure.

FIG. 4 is a cross sectional view of conductive structures of a semiconductor device according to an embodiment of the disclosure.

FIG. 5 is a diagram showing a multi-die device according to an embodiment of the disclosure.

FIG. 6 is a schematic diagram of semiconductor devices coupled together according to an embodiment of the disclosure.

FIG. 7 is a plan diagram of bond pads of semiconductor devices coupled together according to an embodiment of the disclosure.

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FIG. 8 is a flow diagram for providing a signal to semiconductor devices of a stack of semiconductor devices according to an embodiment of the disclosure.

DETAILED DESCRIPTION

Apparatuses and methods for coupling semiconductor devices are disclosed. In a group of semiconductor devices (e.g., a stack of semiconductor devices), a signal is provided to a point of coupling at an intermediate semiconductor device of the group, and the signal is propagated away from the point of coupling over different (e.g., opposite) signal paths to other semiconductor devices of the group. Loading from the point of coupling at the intermediate semiconductor device to other semiconductor devices of a group may be more balanced than, for example, having a point of coupling at semiconductor device at an end of the group (e.g., a lowest semiconductor device of a stack, a highest semiconductor device of the stack, etc.) and providing a signal therefrom. The more balanced topology may reduce a timing difference between when signals arrive at each of the semiconductor devices.

Various embodiments of the present disclosure will be explained below in detail with reference to the accompanying drawings. The following detailed description refers to the accompanying drawings that show, by way of illustration, specific aspects and embodiments of the disclosure. The detailed description includes sufficient detail to enable those skilled in the art to practice the embodiments of the disclosure. Other embodiments may be utilized, and structural, logical and electrical changes may be made without departing from the scope of the present disclosure. The various embodiments disclosed herein are not necessary mutually exclusive, as some disclosed embodiments can be combined with one or more other disclosed embodiments to form new embodiments.

FIG. 1 is a diagram showing a multi-die device **100**. The multi-die device **100** may include a stack **115** of semiconductor devices **120**. The stack **115** is shown as including 10 semiconductor devices **120(0)**-**120(9)**. Each of the semiconductor devices **120(0)**-**120(9)** may be, for example, a memory device, such as a dynamic random access memory (DRAM) device.

The semiconductor devices **120** of the stack **115** may be attached to one another, and the stack **115** may be attached to a substrate **130**. The substrate **130** may include conductive signal lines to route signals along the substrate to and from the stack **115** of semiconductor devices **120**. One or more of the signal lines may be coupled to external terminals **140**, such as solder balls, or the like. Other circuits may also be attached to the substrate **130** and coupled to the conductive signal lines as well. As a result, the circuits attached to the substrate **130** may be coupled, for example, to the stack **115** of semiconductor devices **120** through the conductive signal lines. Circuit **135** attached to the substrate **130** and coupled to the stack **115** of semiconductor devices **120** is shown in FIG. 1. The circuit **135** may be a register clock driver (RCD). The RCD circuit receives and buffers signals provided to the multi-die device **100** and may provide the signals to the stack **115** of semiconductor devices **120**.

The semiconductor devices **120** may be stacked in a staggered manner, providing a “shingle-stack” configuration for the stack **115**. That is, the semiconductor devices **120** are offset from one another to allow edge regions of the semiconductor devices **120** to be exposed. The exposed edge regions may include bond pads to which conductors **125** are

coupled. The conductors **125** may be bond wires, and the bond pads of the edge regions may be coupled together by the bond wires.

Except for the lowest elevation semiconductor device **120(0)**, bond pads of each semiconductor device **120** are coupled to bond pads of an immediately preceding semiconductor device **120** in the stack. For example, as shown in FIG. **1**, a bond pad of semiconductor die **120(9)** is coupled to a bond pad of semiconductor device **120(8)** through bond wire **125(9)**, a bond pad of semiconductor device **120(8)** is coupled to a bond pad of semiconductor device **120(7)** through bond wire **125(8)**, a bond pad of semiconductor device **120(7)** is coupled to a bond pad of semiconductor device **120(6)** through bond wire **125(7)**, and so on down to a bond pad of semiconductor device **120(0)**. The semiconductor device **120(0)** is coupled to the conductive signal lines of the substrate **130** through bond wire **125(0)**. As a result, circuits attached to the substrate **130** (e.g., circuit **135**) may be coupled, for example, to the semiconductor devices **120** through the conductive signal lines and the bond wires **125**.

With the arrangement shown in FIG. **1**, the arrival time of a signal provided to the stack **115** (e.g., from circuit **135**) is different for the semiconductor devices **120** due to the inherent propagation delay through the bond wires **125**. For example, a signal first arrives at the semiconductor device **120(0)** through the bond wire **125(0)**, then arrives at the semiconductor device **120(1)** at a later time after propagating through the bond wire **125(1)**, then arrives at the semiconductor **120(2)** at yet a later time after propagating through the bond wire **125(2)**, and so on, until the signal finally arrives at the semiconductor **120(9)** after propagating through the bond wire **125(9)**. As the number of semiconductor devices **120** are included in a stack, and as memory speeds continue to increase, the time delay between the arrival time of a signal at a first one of the semiconductor devices **120** of the stack **115** and a last one of the semiconductor devices **120** may have greater impact on the performance of the multi-die device **100**.

FIG. **2** is a block diagram of a semiconductor device **210** in accordance with an embodiment of the present disclosure. The semiconductor device **210** may be a memory integrated into a single semiconductor chip (e.g., a semiconductor die) in some embodiments, for example. Example memories may include volatile memory, such as dynamic random access memories (DRAMs) and static random access memories (SRAMs), and non-volatile memories, such as flash memory, resistive memories, and ferroelectric memories or any combination of the above.

The semiconductor device **210** includes a memory cell array **211**. The memory cell array **211** includes a plurality of banks, each bank including a plurality of word lines WL, a plurality of bit lines BL, and a plurality of memory cells MC arranged at intersections of the plurality of word lines WL and the plurality of bit lines BL. The selection of the word line WL is performed by a row decoder/driver **212** and the selection of the bit line BL is performed by a column decoder/driver **213**. Sense amplifiers **218** are coupled to corresponding bit lines BL and connected to local I/O line pairs LIOT/B. Local IO line pairs LIOT/B are connected to main IO line pairs MIOT/B via transfer gates TG **219** which function as switches.

The semiconductor device **210** includes a plurality of terminals. In some embodiments of the disclosure, the terminals may be die pads. The plurality of terminals includes command and address terminals **221**, clock terminals **223** and **223'**, data terminals **224**, a data strobe terminal

224', power supply terminals **225** and **226**. The data terminals **224** may be coupled to output buffers of an input/output circuit **217** for read operations. Alternatively, the data terminals **224** may be coupled to input buffers of the input/output circuit **217** for write access of the memories responsive to a data strobe signal DQS provided at the data strobe terminal **224'**.

The command and address terminals **221** are supplied with command and address signals CA, which include commands and memory addresses. The addresses provided to the command and address terminals **221** are transferred via a command and address input circuit **231** to an address decoder **232**. The address decoder **232** receives the addresses and supplies a decoded row address to the row decoder/driver **212**, and a decoded column address to the column decoder/driver **213**. The address decoder **232** also receives a bank address and provides the bank address to the row decoder/driver **212** and/or the column decoder/driver **213**. In a self-refresh mode, a self-refresh circuit **238** may provide a row address to the row/decoder driver **212** for self-refresh operation.

Commands provided to the command and address terminals **221** are provided to a command decoder **234** via the command and address input circuit **231**. The command decoder **234** decodes the commands and provides the decoded commands to an internal control signal generator **237**. The internal control signal generator **237** may generate various internal command signals responsive to the decoded commands from the command decoder **234**. For example, the internal commands may include a row command signal to select a word line and a column command signal, such as a read command or a write command, to select a bit line.

Accordingly, when an activation command is issued and a row address is timely supplied with the activation command, and a column address is timely supplied with a read command, read data is read from a memory cell MC in the memory cell array **211** designated by the row address and column address. The read data DQ is output from the data terminals **224** via a read/write amplifier **215** and an input/output circuit **217**. Similarly, when an activation command is issued and a row address is timely supplied with the activation command, and a column address is timely supplied with a write command, write data DQ supplied to the data terminals **224** is supplied via the input/output circuit **217** and the read/write amplifier **215** to the memory cell array **211** and written in the memory cell MC designated by the row address and the column address.

The clock terminals **223** are supplied with clock signals CK_t and CK_c, respectively, and clock terminal **223'** are supplied with data clock signals WCK_t and WCK_c. The clock input circuit **235** receives the clock signals CK_t and CK_c and WCK_t and WCK_c and generates internal clock signals ICLK. The internal clock signals ICLK may include internal clock signals based on the clock signals CK_t and CK_c and/or internal clock signals based on the data clock signals WCK_t and WCK_c. Some or all of the internal clock signals ICLK are supplied to an internal clock and timing generator **236**, which generates phase controlled internal clock signals LCLK in response. Although not limited thereto, the internal clock and timing generator **236** may include a DLL circuit. The phase controlled internal clock signals LCLK may be supplied to the input/output circuit **217** and used for timing the input and/or output of data. The internal clock and timing generator **236** may further generate various other internal clock signals for various memory operations.

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The power supply terminals **225** are supplied with power supply potentials VDD and VSS. These power supply potentials VDD and VSS are supplied to a power circuit **239**. The power circuit **239** may generate various internal potentials, for example, VPP, VOD, VARY, VPERI, and the like. The internal potential VPP is mainly used in the row decoder/driver **212**, the internal potentials VOD and VARY are mainly used in the sense amplifiers **218** included in the memory cell array **211**, and the internal potential VPERI is used in many other circuit blocks. The power supply terminals **226** are supplied with power supply potentials VDDQ and VSSQ. These power supply potentials VDDQ and VSSQ are supplied to the input/output circuit **217**. The power supply potentials VDDQ and VSSQ may be the same potentials as the power supply potentials VDD and VSS that are supplied to the power supply terminals **225**, respectively. However, the power supply potentials VDDQ and VSSQ may be used for the input/output circuit **217** so that power supply noise generated by the input/output circuit **217** does not propagate to the other circuit blocks.

FIG. 3 is a layout diagram of a semiconductor device **310** in accordance with one embodiment of the present disclosure. In some embodiments of the disclosure, the semiconductor device **310** includes the semiconductor device **210** of FIG. 2.

The semiconductor device **310** may have edges **350a**, **350b**, **350c** and **350d**, which define ends of the semiconductor device **310**. The edges **350b** and **350d** may extend along a first direction **357a** and the edges **350a** and **350c** may extend along a second direction **357b**, which may be perpendicular to the first direction **357a**. The semiconductor device **310** may include a pad formation area **351**, a peripheral circuit area **352**, and memory cell array areas **353** that include memory cells, circuits, and signal lines, for example, sense amplifier circuits, address decoder circuits, data input/output lines, etc. The peripheral circuit area **352** may include various circuits and signal lines for performing various operations of the semiconductor device **310**. For example, the peripheral circuit area **352** may include command and address input circuits, address and command decoders, clock circuits, power circuits, and input/output circuits. The peripheral circuit area **352** may also include terminals coupled to various circuits of the semiconductor device (e.g., as previously described for the semiconductor devices **170** of FIG. 1). The terminals may be, for example, die pads of the semiconductor device.

The pad formation area **351** may include a plurality of bond pads **354** disposed along the edge **350a**. The plurality of bond pads **354** may be coupled to the terminals (e.g., die pads) of the semiconductor device and represent external terminals of the semiconductor device **310**. For example, the plurality of bond pads **354** may include data terminals, command and address terminals, clock terminals, and/or power supply terminals.

Circuits included in the memory cell array area **353** and/or circuits of the peripheral circuit area **352** may be coupled to one or more bond pads **354** included in the pad formation area **351**. As previously described, various circuits of the semiconductor device **300** may be coupled to the terminals (e.g., die pads). Conductive structures may be used to couple the terminals to one or more of the bond pads **354**. As a result, the circuits coupled to the terminals are also coupled to the bond pads **354**. The conductive structures may extend from locations of the terminals included in the memory cell array area **353** and/or the peripheral circuit area **352** to the

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pad formation area **351**. In some embodiments of the disclosure, the conductive structures may include conductive redistribution layers (RDLs).

FIG. 4 is a cross sectional view of conductive structures of a semiconductor device according to an embodiment of the disclosure. In some embodiments of the disclosure, the conductive structure may be included in the semiconductor device **210** of FIG. 2, the semiconductor devices **120** of the multi-die device **100** of FIG. 1, the semiconductor device **310** of FIG. 3, and/or any semiconductor device according to an embodiment of the disclosure. For example, in some embodiments of the disclosure, the conductive structure may be included in a conductive redistribution layer (RDL) in one or more of the semiconductor devices of FIGS. 1-3.

A conductive structure **430** may couple a terminal **440** of a semiconductor device to a bond pad **420**. The terminal **440** may be a die pad in some embodiments of the disclosure. The terminal **440** may be coupled to one or more circuits **445** of the semiconductor device. As a result, the bond pad **420** may be coupled to the circuits **445** through the conductive structure **430** and the terminal **440**. The circuits **445** may be, for example, circuits included in a memory cell array area (e.g., memory cell array area **352** of FIG. 3) in some embodiments of the disclosure. The circuits **445** may be, for example, circuits included in a peripheral circuit area (e.g., peripheral circuit area **354** of FIG. 3) in some embodiments of the disclosure. The circuits **445** may be for performing various operations of the semiconductor device. In some embodiments of the disclosure, the circuits **445** may include command and address input circuits, address and command decoders, clock circuits, power circuits, and input/output circuits, as well as other circuits.

The bond pad **420**, conductive structure **430**, terminal **440**, and circuits **445** of the semiconductor device may be formed from semiconductor structures. The semiconductor structures may include conductive layers, conductive vias, insulating interlayers, etc. The terminal **440** may be coupled to the circuits **445** through, for example, metal layers and/or conductive vias. The conductive structure **430** may be disposed on one or more insulating interlayers. As previously described, the conductive structure **430** may be further coupled to the bond pad **420**. The conductive structure **430** extends from the terminal **440** to the bond pad **420** so that the bond pad **420** and the circuit terminal **440** may be coupled together. As a result, a circuit **445** coupled to the terminal **440** may be externally accessible through the bond pad **420**. The bond pad **420** may be exposed by an opening **425** in a passivation layer **435**. In some embodiments of the disclosure, the passivation layer **435** may be a polyimide material.

In some embodiments of the disclosure, the conductive structure **430** may be coupled to the terminal **440** of the semiconductor device (e.g., die pad of the semiconductor device) to provide a bond pad **420** for coupling to a terminal **440** in a different location. For example, the terminal **440** may represent die pads that are generally located in a peripheral circuit region along a center region of the semiconductor device, and the conductive structure **430** may couple the die pads in the center region to bond pads **420** that are located along an edge of the semiconductor device. The bond pads **420** along an edge may be more conveniently located and provide coupling to circuits **445** of the semiconductor device, that is, through the conductive structure **430** and the terminal **440**.

FIG. 5 is a diagram showing a multi-die device **500** according to an embodiment of the disclosure. The multi-die device **500** may include a stack **515** of semiconductor

devices **520**. Embodiments of the disclosure are not limited to the particular number of semiconductor devices **520** included in the stack **515** shown in FIG. **5**. In some embodiments of the disclosure, each of the semiconductor devices **520** includes the semiconductor device **210** of FIG. **2**. Each of the semiconductor devices **520** may be, for example, a memory device, such as a dynamic random access memory (DRAM) device.

The semiconductor devices **520** of the stack **515** may be attached to one another, and the stack **515** may be attached to a substrate **530**. In some embodiments of the disclosure, the semiconductor devices **520** are attached to one another and/or to the substrate **530** by an adhesive epoxy. The substrate **530** may include conductive signal lines to route signals along the substrate to and from the stack **515** of semiconductor devices **520**. One or more of the signal lines may be coupled to external terminals **540**. In some embodiments of the disclosure, the external terminals **540** may include solder balls, or the like. Other circuits may also be attached to the substrate **530** and coupled to the conductive signal lines as well. As a result, the circuits attached to the substrate **530** may be coupled, for example, to the stack **515** of semiconductor devices **520** through the conductive signal lines. Circuit **535** attached to the substrate **530** and coupled to the stack **515** of semiconductor devices **520** is shown in FIG. **5**. The circuit **535** may be a register clock driver (RCD) in some embodiments of the disclosure. In such embodiments of the disclosure, the RCD circuit receives and buffers signals provided to the multi-die device **500** and may provide the signals to the stack **515** of semiconductor devices **520**. Additional or alternative circuits may be included in the multi-die device **500**, and/or the circuit **535** may be other circuits as well without departing from the scope of the disclosure.

The semiconductor devices **520** may be stacked in a staggered manner, providing a “shingle-stack” configuration for the stack **515**. That is, the semiconductor devices **520** are offset from one another to allow edge regions of the semiconductor devices **520** to be exposed. The exposed edge regions may include bond pads to which conductors **525** are coupled. In some embodiments of the disclosure, the conductors **525** may be bond wires, and the bond pads of the edge regions may be coupled together by the bond wires.

Except for the lowest elevation semiconductor device **520(0)**, bond pads of each semiconductor device **520** are coupled to bond pads of at least one other semiconductor device **520** in the stack. For example, as shown in FIG. **5**, a bond pad of semiconductor device **520(8)** is coupled to a bond pad of semiconductor device **520(9)** through conductor **525(9)**, and the bond pad is further coupled to a bond pad of semiconductor device **520(7)** through conductor **525(8)**. Similarly, a bond pad of semiconductor device **520(1)** is coupled to a bond pad of semiconductor device **520(2)** through conductor **525(2)**, and the bond pad is further coupled to a bond pad of semiconductor device **520(0)** through conductor **525(1)**.

The semiconductor device **520(4)** is coupled to the conductive signal lines of the substrate **530** through source conductor **510**. The source conductor **510** may be coupled, for example, to a bond pad the semiconductor device **520(4)**. The semiconductor device **520(4)** is coupled to semiconductor device **520(3)**, which is coupled to semiconductor device **520(2)**, which is coupled to semiconductor device **520(1)**, which is coupled to semiconductor device **520(0)**. The semiconductor device **520(4)** is also coupled to the semiconductor device **520(5)**, which is coupled to the semiconductor device **520(6)**, which is coupled to the semicon-

ductor device **520(7)**, which is coupled to the semiconductor device **520(8)**, which is coupled to the semiconductor device **520(9)**. As a result, circuits attached to the substrate **530** (e.g., circuit **535**) may be coupled, for example, to the stack **515** of semiconductor devices **520** through the conductive signal lines and the source conductor **510**, and the conductors **525**.

In embodiments of the disclosure, coupling to the stack **515** at an intermediate semiconductor device of the semiconductor devices **520** may reduce the difference in timing between when signals arrive at each of the semiconductor devices. An intermediate semiconductor device may be any semiconductor device between a lowest semiconductor device and a highest semiconductor device (e.g., between semiconductor device **520(0)** (lowest) and semiconductor device **520(9)** (highest)). Coupling to an intermediate semiconductor device of the stack **515** may provide a more balanced topology from a point of coupling.

While FIG. **5** illustrates the semiconductor devices **520** in a stack **515**, in other embodiments of the disclosure, the semiconductor devices **520** may be arranged differently in a group. For example, the semiconductor devices may be arranged lined up laterally relative to one another. The point of coupling may be to an intermediate semiconductor device of the group of laterally arranged semiconductor devices. In such embodiments, a signal provided to the point of coupling may be propagated laterally in opposite directions from the point of coupling to semiconductor devices at opposite ends of the lateral arrangement of semiconductor devices. In some embodiments of the disclosure, the semiconductor devices **520** are physically positioned relative to one another in other arrangements. More generally, the scope of the disclosure is not intended to be limited to only a stacked arrangement of semiconductor devices.

For example, in the embodiment of the disclosure shown in FIG. **5**, conductive signal lines of the substrate **530** are coupled to the stack **515** at intermediate semiconductor device **520(4)** through source conductor **510**. Loading from the point of coupling at the semiconductor device **520(4)** to the lowest semiconductor device **520(0)** and to the highest semiconductor device **520(9)** is more balanced than, for example, having a point of coupling at the lowest semiconductor device. As a result, the difference in the timing of signals from the point of coupling at the semiconductor device **520(4)** to the lowest semiconductor device **520(0)** and to the highest semiconductor device **520(9)** of the stack may be decreased.

It may be advantageous to have the point of coupling at an intermediate semiconductor device near a center of a group (e.g., stack) of semiconductor devices. A “center” semiconductor device is an intermediate semiconductor device of a group that may be positioned in the group so there are an equal number of semiconductor devices in either direction from the center semiconductor device to opposite end semiconductor devices. For example, in a stack of nine semiconductor devices, the center semiconductor device is the fifth semiconductor device, which is positioned in the stack so there are four semiconductor devices in a first direction to a first end semiconductor device (e.g., the lowest semiconductor device) and in a second direction to a second end semiconductor device (e.g., the highest semiconductor device). Where a group includes an even number of semiconductor devices, the center semiconductor device may be either of two semiconductor devices that are positioned in the group so there are almost an equal number of semiconductor devices (e.g., +/-1 semiconductor device) in either direction from the center semiconductor device to opposite

end semiconductor devices. For example, in a stack of eight semiconductor devices, the center semiconductor device is either the fourth or fifth semiconductor device, which is positioned in the stack so there are four semiconductor devices in a first direction to a first end semiconductor device and three semiconductor devices in a second direction to a second end semiconductor device.

FIG. 6 is a schematic diagram of semiconductor devices coupled together according to an embodiment of the disclosure. In some embodiments of the disclosure, the semiconductor devices of FIG. 6 may represent the stack 515 of semiconductor devices 520 of FIG. 5. In some embodiments of the disclosure, the semiconductor devices of FIG. 6 may represent a group of semiconductor devices arranged laterally relative to one another. In some embodiments of the disclosure, the semiconductor devices of FIG. 6 may represent the semiconductor devices in other physical arrangements. For the purpose of ease of description, the semiconductor devices of FIG. 6 will be described with reference to the semiconductor devices 520 of FIG. 5. However, the semiconductor devices of FIG. 6 are not necessarily limited to such.

Bond pads of the semiconductor devices 520(0)-520(9) are coupled by conductors 525(1)-525(9). For example, a bond pad of semiconductor device 520(0) is coupled to a bond pad of semiconductor device 520(1) by conductor 525(1); the bond pad of semiconductor device 520(1) is coupled to a bond pad of semiconductor device 520(2) by conductor 525(2); the bond pad of semiconductor device 520(2) is coupled to a bond pad of semiconductor device 520(3) by conductor 525(3); the bond pad of semiconductor device 520(3) is coupled to a bond pad of semiconductor device 520(4) by conductor 525(4); the bond pad of semiconductor device 520(4) is coupled to a bond pad of semiconductor device 520(5) by conductor 525(5); the bond pad of semiconductor device 520(5) is coupled to a bond pad of semiconductor device 520(6) by conductor 525(6); the bond pad of semiconductor device 520(6) is coupled to a bond pad of semiconductor device 520(7) by conductor 525(7); the bond pad of semiconductor device 520(7) is coupled to a bond pad of semiconductor device 520(8) by conductor 525(8); and the bond pad of semiconductor device 520(8) is coupled to a bond pad of semiconductor device 520(9) by conductor 525(9).

A source conductor 510 is coupled to the bond pad of the semiconductor device 520(4). The source conductor 510 may be coupled to another circuit and/or conductive signal lines, for example, of a substrate. As a result, the circuit and/or conductive signal lines may be coupled to the other semiconductor devices.

In the example of FIG. 6, only one bond pad is shown for each of the semiconductor devices 520. However, each of the semiconductor devices 520 may include additional bond pads that may be coupled to other semiconductor devices 520 without departing from the scope of the disclosure. The bond pads of the semiconductor devices 520 may also represent one or more physical bonds pads that may be coupled together, and thus, are not limited to merely one physical bond pad to which multiple conductors are coupled. For example, in some embodiments of the disclosure, the bond pads of semiconductor devices may represent two bond pads coupled together, one bond pad coupled to a bond pad of a first adjacent semiconductor device and another bond pad coupled to a bond pad of a second adjacent semiconductor device. In some embodiments of the disclosure, the bond pads of semiconductor devices may represent three bond pads coupled together, a first bond pad coupled

to a bond pad of a first adjacent semiconductor device, a second bond pad coupled to a bond pad of a second adjacent semiconductor device, and a third bond pad to which a source conductor 510 may be coupled.

The conductors 525 include inherent propagation delays which can affect the timing of when signals arrive at the respective semiconductor device. However, with the source conductor 510 coupled to an intermediate one of the semiconductor devices 520(0)-520(9), a more balanced topology may be provided. An intermediate semiconductor device may be, for example, a semiconductor device between a first end semiconductor device and a second end semiconductor device (e.g., between semiconductor device 520(0) (first end) and semiconductor device 520(9) (second end)). The more balanced topology may reduce a timing difference between when signals arrive at each of the semiconductor devices 520. For example, as shown in the example of FIG. 6, with the point of coupling at the bond pad of the semiconductor device 520(4), a signal provided on the source conductor 510 propagates through four conductors 525(4)-525(1) to arrive at the semiconductor device 520(0) (e.g., first end semiconductor device) and propagates through five conductors 525(5)-525(9) to arrive at the semiconductor device 520(9) (e.g., second end semiconductor device).

In an embodiment of the disclosure where each of the conductors 525 provide a similar propagation delay, the difference in arrival time of a signal provided on the source conductor 510 to the semiconductor device 520(0) and to the semiconductor device 520(9) is approximately the propagation delay of one conductor 525 (e.g., conductor 525(9), the fifth conductor), which may be considered in some embodiments of the disclosure to be approximately the same propagation delay to the semiconductor device 520(0) and to the semiconductor device 520(9).

In contrast, the difference in signal timing to an arrangement where a signal is first provided to semiconductor device 520(0) (e.g., a first end semiconductor device) and the signal propagates through conductors 525(1)-525(9) to the semiconductor device 520(9) may be significantly greater. For example, the signal arrives at the semiconductor device 520(9) after the propagation delays of nine conductors 525. Thus, the difference in signal timing between the semiconductor device 520(0) and the semiconductor device 520(9) is the total propagation delay of nine conductors 525, which may be a significantly longer delay than a timing difference from the propagation delay of one conductor 525, such as for the example of FIG. 6.

An arrangement of coupling a source conductor to an intermediate semiconductor device of a group of semiconductor devices (e.g., a stack of semiconductor devices) may be advantageous in reducing a timing difference of signal arrival at two end semiconductor devices. Such an arrangement may provide improved signal integrity for uni-directional signals with high loading, for example, command and address signals, chip select signals, clocks, as well as other signals. An arrangement as disclosed may also require lighter drive and lighter terminal termination (e.g., on die termination) to provide valid data eyes at a receiver circuit.

FIG. 7 is a plan diagram of bond pads of semiconductor devices coupled together according to an embodiment of the disclosure. While FIG. 7 shows five semiconductor devices 720(2)-720(6), additional semiconductor devices may be coupled to the semiconductor devices 720(2)-720(6) without departing from the scope of disclosure. For example, in some embodiments of the disclosure, the semi-

conductor devices may be included in a multi-die device (e.g., multi-die device **200** of FIG. **2**) in some embodiments of the disclosure.

The bond pads **715-717** may be included in a pad formation area of respective semiconductor device **720(2)-720(6)**. Corresponding bond pads of each semiconductor device are coupled together through conductors **725**. For example, the bond pads **715(2)-715(6)** of the semiconductor devices **720(2)-720(6)** are coupled together through conductors **725(3)A-725(6)A**; the bond pads **716(2)-716(6)** of the semiconductor devices **720(2)-720(6)** are coupled together through conductors **725(3)B-725(6)B**; and the bond pads **717(2)-717(6)** of the semiconductor devices **720(2)-720(6)** are coupled together through conductors **725(3)C-725(6)C**. While only three bond pads **715-717** are shown for each semiconductor device **720**, additional bond pads may be included without departing from the scope of the disclosure.

A respective conductor **710A-710C** is coupled to bond pads **715(4)-717(4)** of intermediate semiconductor device **720(4)**. Each of the conductors **710A-710C** may be coupled to circuits and/or conductive signals lines, for example, of a substrate. Signals provided on the respective conductors **710** may be provided to each of the semiconductor devices through the conductors **725** coupled to the respective bond pads **715-717**. The signal timing for the signals may be more balanced by having the point of coupling at an intermediate semiconductor device, and propagating the signal away from the point of coupling to semiconductor devices on two different (e.g., opposite) signal paths through the conductors. For example, the signal timing of a signal arriving at the semiconductor device **720(2)** and arriving at the semiconductor device **720(6)** may be approximately equal because the propagation delay through the conductors **725** should be approximately equal from the point of coupling at the bond pads **715(4)-717(4)** of the semiconductor device **720(4)** to the corresponding bond pads **715(2)-717(2)** of the semiconductor device **720(2)**, and from the point of coupling at the bond pads **715(4)-717(4)** to the corresponding bond pads **715(6)-717(6)** of the semiconductor device **720(6)**.

FIG. **8** is a flow diagram of providing a signal to semiconductor devices of a stack of semiconductor devices according to an embodiment of the disclosure. In some embodiments of the disclosures, the semiconductor devices may be included in the stack of semiconductor devices of FIG. **5** and/or the coupled semiconductor devices of FIG. **6**. In some embodiments of the disclosure, the semiconductor devices of the stack for the method of FIG. **8** may include bond pads coupled together as described with reference to FIG. **7**. The semiconductor devices are memory devices, for example, DRAM devices in some embodiments of the disclosure. For the purpose of ease of description, a method **800** of the flow diagram of FIG. **8** will be described with reference to stack **515** of semiconductor devices **520** of FIG. **5**. However, the method **800** of FIG. **8** is not necessarily limited to such.

At step **801** a signal is provided to an intermediate semiconductor device of a stack of semiconductor devices. The intermediate semiconductor device may be between a first semiconductor device and a second semiconductor device. For example, with reference to the stack **515** of semiconductor devices **520** of FIG. **5**, the semiconductor device **520(4)** may be an intermediate semiconductor device, which is between semiconductor devices **520(3)** and **520(5)**. The signal may, for example, be a signal provided by a circuit external to the stack, such as circuit **535**. As previously described, the circuit **535** and the stack **515** may be attached to a substrate that includes conductive signal

lines. The signal may be provided to the semiconductor device **520(4)** through the conductive signal lines and by source conductor **510** coupled to a conductive signal line and coupled to a bond pad of the semiconductor device **520(4)**.

At step **803** the signal is provided away from the intermediate semiconductor device to the first semiconductor device on a first signal path. For example, again with reference to the stack **515** of semiconductor devices **520** of FIG. **5**, the semiconductor device **520(3)** may be the first semiconductor device and the first signal path may be conductor **525(3)**. The conductor **525(3)** may be coupled to the bond pad of the semiconductor device **520(4)** and to a bond pad of the semiconductor device **520(3)**. The signal provided to the semiconductor device **520(4)** (e.g., to the bond pad of the semiconductor device **520(4)**) may be provided away from the semiconductor device **520(4)** to the semiconductor device **520(3)** on the conductor **525(3)**.

At step **805** the signal is also provided away from the intermediate semiconductor device to the second semiconductor device on a second signal path. For example, again with reference to the stack **515** of semiconductor devices **520** of FIG. **5**, the semiconductor device **520(5)** may be the second semiconductor device and the second signal path may be conductor **525(5)**. The conductor **525(5)** may be coupled to the bond pad of the semiconductor device **520(4)** and to a bond pad of the semiconductor device **520(5)**. The signal provided to the semiconductor device **520(4)** may be provided away from the semiconductor device **520(4)** to the semiconductor device **520(5)** on the conductor **525(5)**. In some embodiments of the disclosure, the second signal path (e.g., conductor **525(5)**) and the first signal path (e.g., conductor **525(3)**) may be opposite signal paths to provide the signal away from the intermediate semiconductor device (e.g., semiconductor device **520(4)**).

Signal paths (e.g., conductors **525(3)** and **525(5)**) include inherent propagation delays which can affect the timing of when signals arrive at the respective semiconductor devices. As previously described, providing a signal to an intermediate one of the semiconductor devices (e.g., intermediate semiconductor device **520(4)**) and providing the signal away from the intermediate semiconductor device to first and second semiconductor devices on respective signal paths (e.g., semiconductor device **520(3)** on conductor **525(3)** and semiconductor device **520(5)** on conductor **525(5)**), may provide a more balanced topology. As a result, a timing difference between when signals arrive at the first and second semiconductor devices may be reduced. For example, in some embodiments of the disclosure, due to the coupling through an intermediate semiconductor device, the propagation delay of the signal through the first signal path to the first semiconductor device may be equal to the propagation delay of the signal through the second signal path to the second semiconductor device.

Certain details have been described to provide a sufficient understanding of examples of the disclosure. However, it will be clear to one having skill in the art that examples of the disclosure may be practiced without these particular details. Moreover, the particular examples of the present disclosure described herein should not be construed to limit the scope of the disclosure to these particular examples. In other instances, well-known circuits, control signals, timing protocols, and software operations have not been shown in detail in order to avoid unnecessarily obscuring the disclosure. Additionally, terms such as “couples” and “coupled” mean that two components may be directly or indirectly

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electrically coupled. Indirectly coupled may imply that two components are coupled through one or more intermediate components.

From the foregoing it will be appreciated that, although specific embodiments of the disclosure have been described herein for purposes of illustration, various modifications may be made without deviating from the spirit and scope of the disclosure. Accordingly, the scope of the disclosure should not be limited any of the specific embodiments described herein.

What is claimed is:

1. A method, comprising:
 - providing a signal to an intermediate semiconductor device of a stack of semiconductor devices, the intermediate semiconductor device disposed between first and second semiconductor devices included in the stack of semiconductor devices;
 - propagating the signal away from the intermediate semiconductor device to the first semiconductor device through a first signal path coupled to the intermediate semiconductor device and the first semiconductor device, wherein the first semiconductor device is limited to receiving the signal via the first signal path coupled to the intermediate semiconductor device and the first semiconductor device; and
 - propagating the signal away from the intermediate semiconductor device to the second semiconductor device through a second signal path coupled to the intermediate semiconductor device and the second semiconductor device, wherein the second semiconductor device is limited to receiving the signal via the second signal path coupled to the intermediate semiconductor device and the second semiconductor device.
2. The method of claim 1, wherein providing the signal to the intermediate semiconductor device comprises providing the signal to a bond pad of the intermediate semiconductor device, and wherein the first signal path and the second signal path are coupled to the bond pad of the intermediate semiconductor device.
3. The method of claim 1, wherein propagating the signal away from the intermediate semiconductor device on the first and second signal paths comprise propagating the signal away from the intermediate semiconductor device on opposite signal paths.
4. The method of claim 1, wherein providing the signal to the intermediate semiconductor device comprises providing the signal from a circuit coupled to the intermediate semiconductor device through at least a conductive signal line of a substrate and a source conductor coupled to the conductive signal line and the intermediate semiconductor device.
5. The method of claim 1, wherein the intermediate semiconductor device comprises a center memory device of the stack of semiconductor devices.
6. The method of claim 1, wherein the signal is propagated to the first semiconductor device with a first signal

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timing, and wherein the signal is propagated to the second semiconductor device with a second signal timing equal to the first signal timing.

7. The method of claim 1, wherein the stack of semiconductor devices comprises an even number of semiconductor devices.

8. The method of claim 1, wherein the stack of semiconductor devices comprises an odd number of semiconductor devices.

9. The method of claim 1, wherein a first propagation delay of the signal to the first semiconductor device is equal to a second propagation delay of the signal to the second semiconductor device.

10. A method, comprising:

- providing a signal to a second memory device of a plurality of memory devices, wherein the second memory device is disposed between a first memory device and a third memory device of the plurality of memory devices;

propagating the signal from the second memory device to the first memory device via a first signal path, wherein the first memory device is limited to receiving the signal via the first signal path; and

propagating the signal from the second memory device to the third memory device via a second signal path, wherein the third memory device is limited to receiving the signal via the second signal path.

11. The method of claim 10, wherein the plurality of memory devices comprises a stack of memory devices.

12. The method of claim 10, wherein the plurality of memory devices comprises a lowest memory device and a highest memory device.

13. The method of claim 10, wherein the first signal path and the second signal path are opposite signal paths.

14. The method of claim 10, wherein the first signal path and the second signal path have a same propagation delay.

15. The method of claim 10, wherein providing the signal to the second memory device comprises providing the signal to a bond pad of the second memory device, and wherein the first signal path and the second signal path are coupled to the bond pad of the second memory device.

16. The method of claim 10, wherein providing the signal to the second memory device comprises providing the signal from a circuit coupled to the second memory device through at least a conductive signal line of a substrate and a source conductor coupled to the conductive signal line and the second memory device.

17. The method of claim 11, wherein the second memory device comprises a center memory device of the stack of memory devices.

18. The method of claim 10, wherein the plurality of memory devices comprises an even number of memory devices.

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